How-To Library

Informatica

Enable Customer Managed Keys for your Organization on Microsoft Azure

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Abstract

This article explains how to create your own native master encryption key for IDMC on Microsoft Azure. The master encryption key is used to encrypt your organization-specific encryption keys. The key that you create is controlled and maintained by you. You can use it to control and restrict access to your organization's data.

Supported Versions

- Intelligent Data Management Cloud July 2026

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Overview

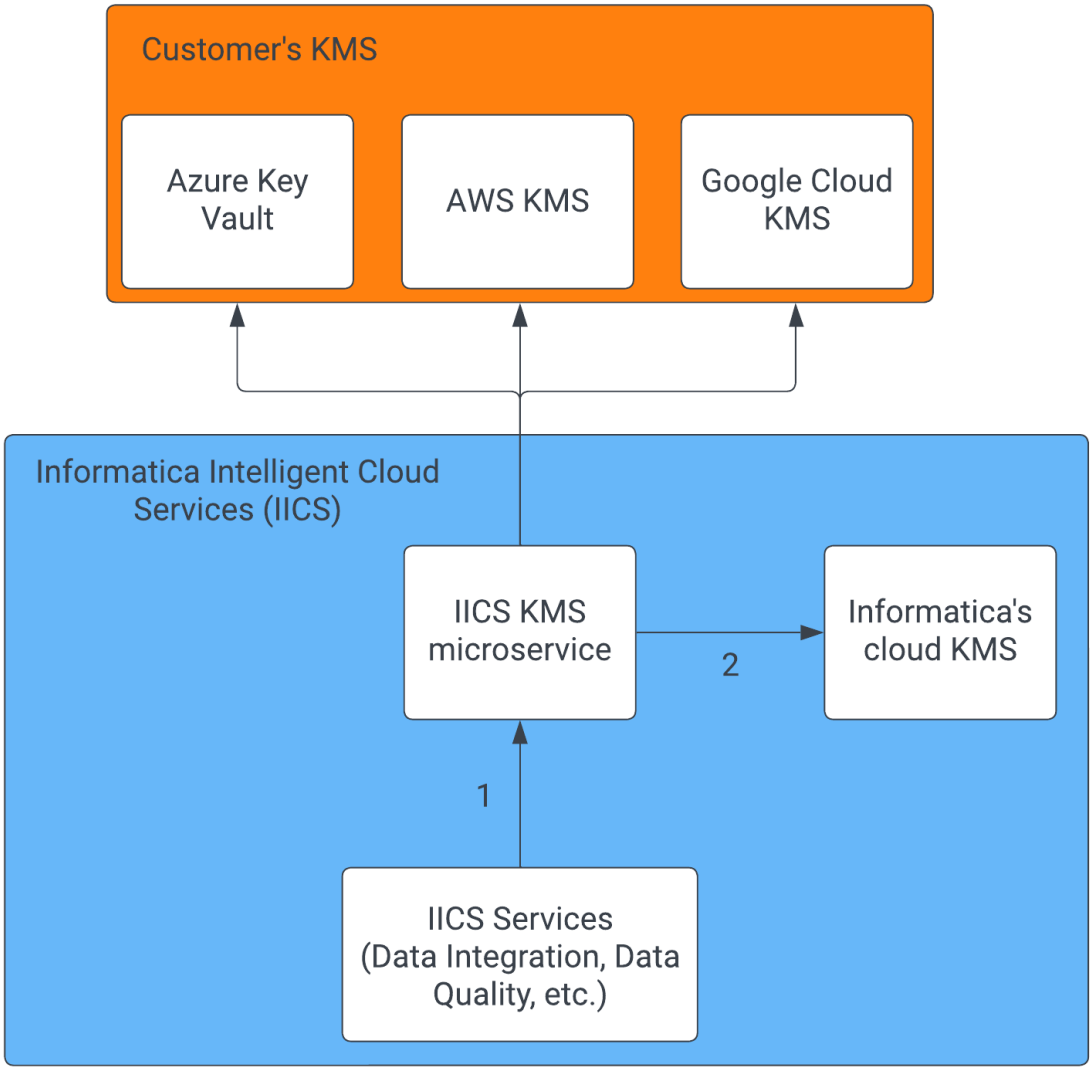
IDMC protects your organization's sensitive data in the cloud using organization-specific encryption keys that are generated and stored in the IDMC key management service (KMS). To prevent malicious access, the keys are encrypted using a master key that is stored in the cloud provider's KMS.

If you prefer, you can create a customer-managed key (CMK). When you create a CMK, you control access to it. However, you'll need to grant IDMC access to the CMK so that it can encrypt and decrypt your organization's sensitive data.

Creating a CMK offers the following benefits:

- You can restrict and control any access to your data.
- You can restrict the decryption of your data in the event of a data breach.
- You create and hold the key material in your KMS. The key is never exposed to your cloud service provider.
- You maintain full control of the key throughout its lifecycle. You can revoke access or delete the key at any time.

The following image shows how IDMC interfaces with your CMK:



1. IDMC interfaces with the IDMC KMS agnostically.
2. Non-customer managed keys go to Informatica's cloud KMS.

Note: When you create a CMK, your KMS and IDMC POD must use the same cloud provider. For example, if your IDMC POD is a Microsoft Azure POD, then you must store your CMK in Azure Key Vault. You can't store it in AWS KMS or Google Cloud KMS.

After you create and enable a CMK, you can revoke it at any time by disabling customer managed keys in IDMC Administrator. If you do this, you'll go back to using Informatica's master key.

Steps for creating and enabling the key

To create and use a CMK, you provision the key in Azure Key Vault and enable cross-account access with IDMC. Then you enable customer managed keys in IDMC.

To create and enable a CMK, complete the following steps:

1. In Azure Key Vault, create a key vault.

2. In the key vault you created, generate the key to use as your CMK.
3. If you access Azure Key Vault from behind a firewall, add Informatica's IP addresses to the vault's allowlist.
4. Authorize Informatica's enterprise Azure application to access the key.
5. Allow Informatica's enterprise application to use the Azure AD application credentials and tenant ID so that it can fetch an access token and make cryptographic calls to the key vault.

Step 1. Create a key vault

In Azure Key Vault, create a key vault to store your CMK. Note the key vault URI because you will need it when you allow Informatica's application to use the key vault.

1. Log in to the Azure portal.
2. In the **Search** box, enter `Key Vault`.
3. From the results list, select **Key Vault**.
4. In the **Key Vault** section, select **Create**.
5. On the **Basics** tab, configure the key vault details and click **Next**.
6. On the **Access configuration** tab, select the permission model, review and update the other access policies as needed, and click **Next**.

   You can choose either the **Azure role-based access control (recommended)** or **Vault access policy** permission model.

   Note: Azure role-based access control (Azure RBAC) is the default and recommended access control model for Azure Key Vault. The Vault access policy model is deprecated and is planned to be retired in February 2027. For more information, see Migrate to Azure RBAC from access policies in the Key Vault documentation.
7. On the **Networking** tab, review the networking details and click **Next**.
8. On the **Tags** tab, add tags to the key vault and click **Next**.
9. On the **Review + create** tab, review the key vault configuration and click **Create**.
10. After deployment completes, click **Go to Resource** to view the key vault you just created and note the vault URI.
11. Under **Objects**, select **Keys** and check your permissions.

    If you see the message "This operation is not allowed by RBAC," perform the following steps to configure your role assignment:

    a. Click **Access control (IAM)**.
    b. On the **Role assignments** tab, click **Add** > **Add role assignment**.
    c. On the **Add role assignment** page, open the **Role** tab, and select the Key Vault Administrator role.
    d. On the **Members** tab, assign the Key Vault Administrator role to yourself or the user that you want to be the key vault administrator.
    e. On the **Review + assign** tab, verify the role assignment settings.
    f. Click **Review + assign**.

    The key vault administrator should be able to create keys at this point.

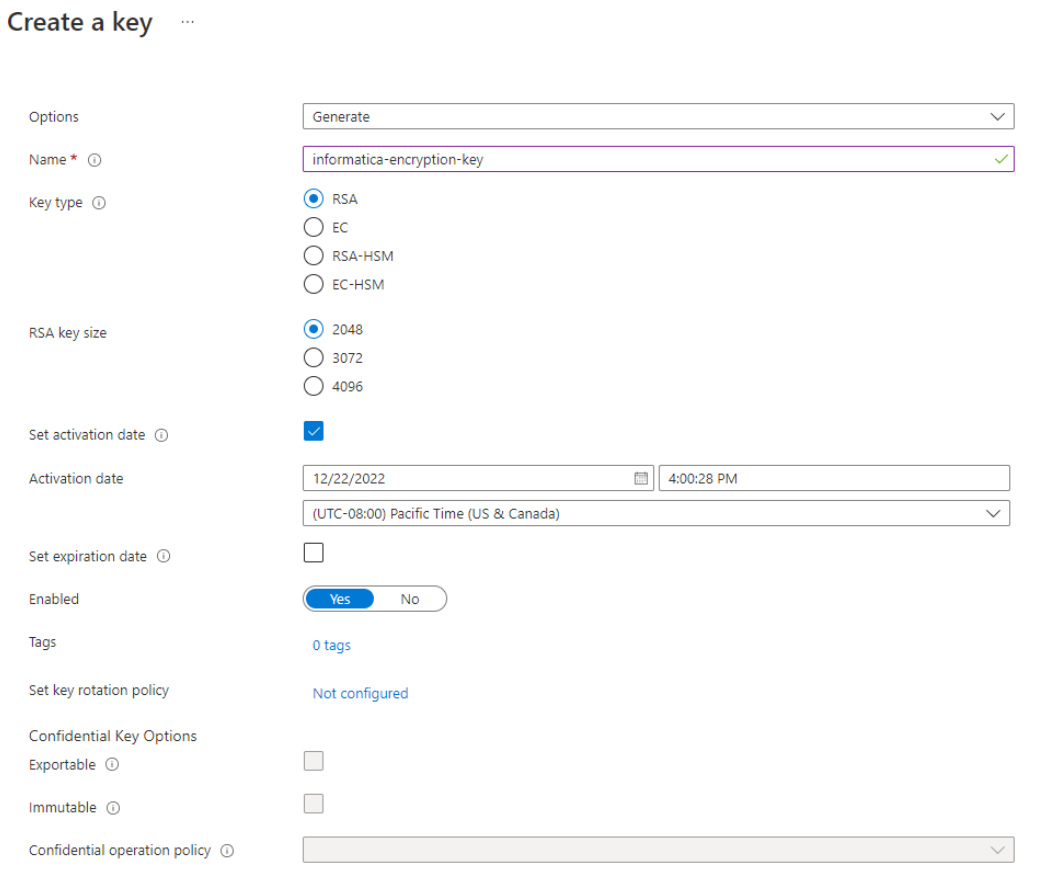
Step 2. Create the key

In the key vault you created, create an asymmetric key to use as your CMK. Note the key name and version because you will need them when you allow Informatica's application to use the key vault.

1. In the **Search** box, enter **Key Vault**.

2. From the results list, select the key vault you created in "Step 1. Create a key vault" on page 4.
3. In the left panel under **Objects**, select **Keys**.
4. Select **Generate/Import**.
5. Select **Generate** from the drop-down list, and configure the following options:

Property	Value
Name	Key name, for example, `informatica-encryption-key`
Key type	RSA
RSA key size	Choose any key size.
Set activation date	Enabled
Activation date	Today's date and time
Enabled	Yes
Set key rotation policy	Optionally, configure a key rotation policy.



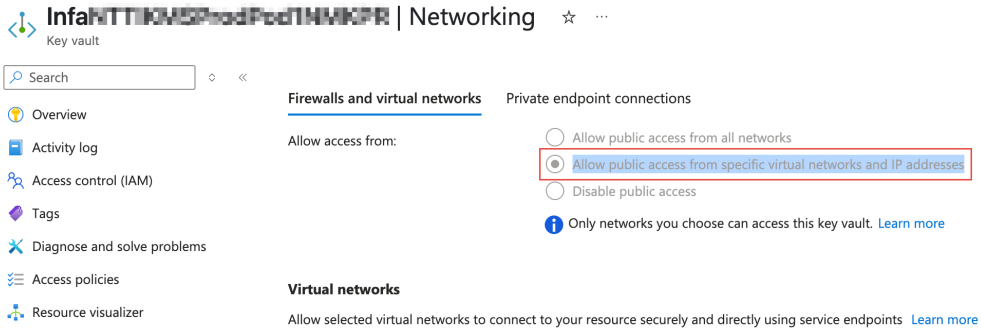
6. Click **Create**.
7. Click the key and note the key name and version.

Step 3. Add Informatica IP addresses to the key vault allowlist

If you access Azure Key Vault from behind a firewall, you'll need to expose the key vault through the public endpoint and add Informatica's IP addresses to the vault's allowlist. If you don't access the key vault from behind a firewall, skip this step.

Note: You need to expose the Key Vault through the public endpoint. IDMC can't connect to the key vault through a private endpoint.

1. In the **Search** box, enter **Key Vault**.
2. From the results list, select the key vault you created in "Step 1. Create a key vault" on page 4.
3. Under **Settings**, select **Networking**, and then select the **Firewalls and virtual networks** tab.
4. Under **Allow access from**, check to see whether **Allow public access from specific virtual networks and IP addresses** is enabled.



5. If this option is enabled, add the primary and disaster recovery IP addresses for your POD to the allowed IP network addresses and save the changes.

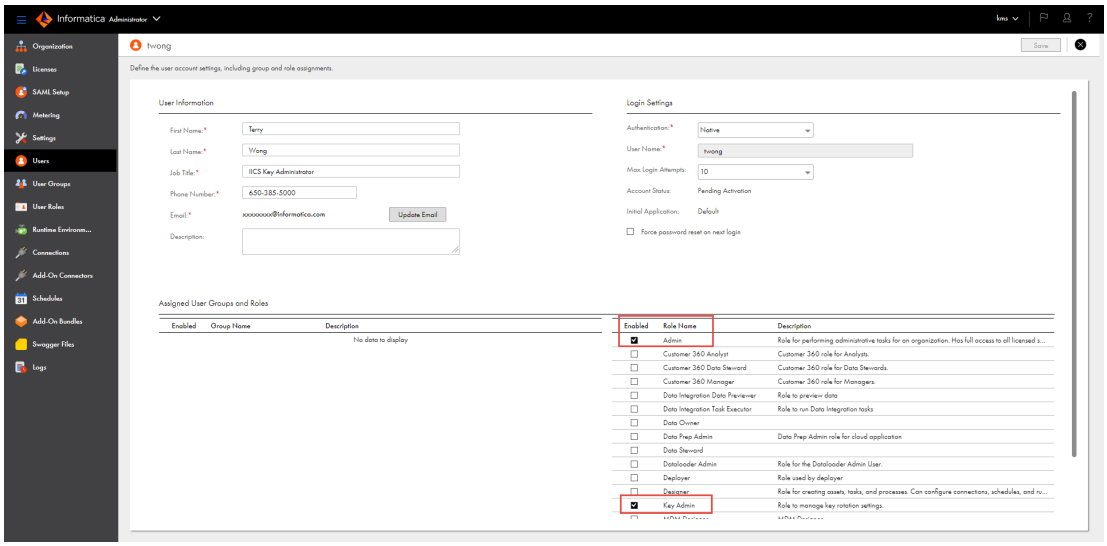
   The IP addresses for your POD are listed on the POD Availability and Networking page.

   To find the IP addresses, click the link for your POD to open the POD information page. Add the primary and disaster recovery IP addresses listed in the **CMK (Customer Managed Key)** row to the allowlist.

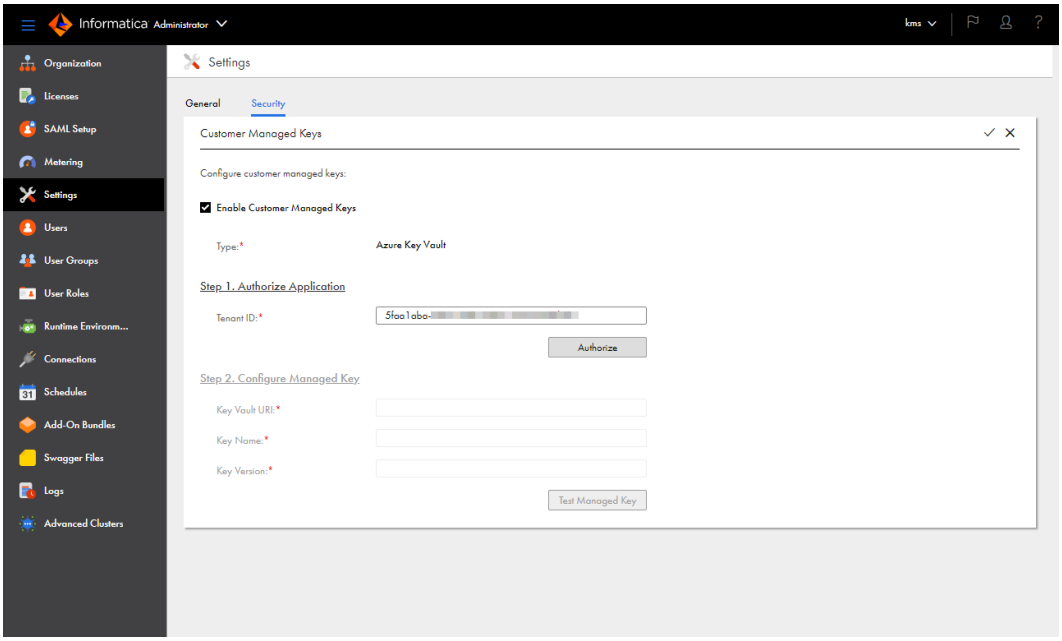
Step 4. Authorize Informatica's enterprise Azure application to access the key

In IDMC Administrator, open the **Settings** page and enable customer managed keys for your organization. Then grant the Informatica KMS Connect app permissions to access your CMK.

Note: Before you can complete this step, you need to assign at least one administrative user the **Admin** and **Key Admin** roles on the user details page in Administrator:



1. Log in to IDMC Administrator with a user account that has both the Admin and Key Admin roles.
2. Open the **Settings** page and click the **Security** tab.
3. Click the edit (pencil) icon.
4. Enable the **Enable Customer Managed Keys** option.
5. In **Step 1. Authorize Application**, enter the **Tenant ID** of your Azure Subscription.



Tip: You can find the Tenant ID under **Properties** in Azure Active Directory.

6. Click **Authorize**.
7. When prompted, sign in to your Azure account.

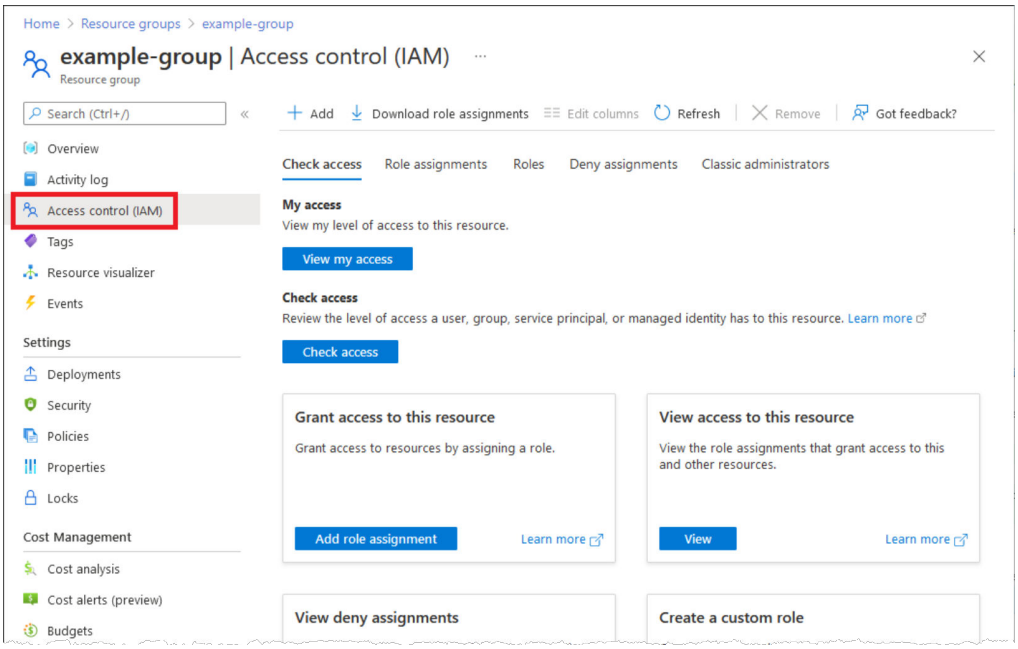
   Note: You must sign in with a user account that has the Cloud Application Administrator role.
8. In the **Permissions requested** dialog, click **Accept** to grant permissions to the Informatica KMS Connect app.

Step 5. Allow Informatica's application to use the key vault

Allow the Informatica KMS Connect app to use the Azure AD application credentials and tenant ID. To do this, create a role assignment or an access policy for the Informatica KMS Connect app in Azure Key Vault, and then enter the key information on the **Settings** page in IDMC.

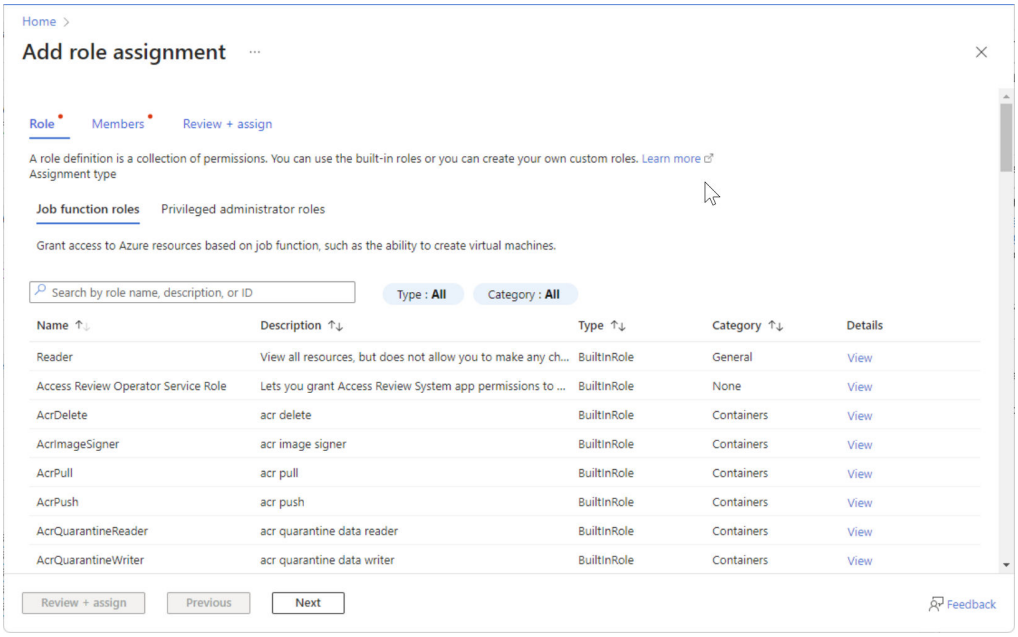
Note: Azure role-based access control (Azure RBAC) is the default and recommended access control model for Azure Key Vault. The Vault access policy model is deprecated and is planned to be retired in February 2027. For more information, see Migrate to Azure RBAC from access policies in the Key Vault documentation.

1. Log in to the Azure portal.
2. In the **Search** box, enter **Key Vault**.
3. From the results list, select **Key Vault**.
4. Select the key vault you created in "Step 1. Create a key vault" on page 4.
5. If you use Azure RBAC, perform the following steps to create a role assignment for the Informatica KMS Connect app:
   a. In the left panel, select **Access control (IAM)**.



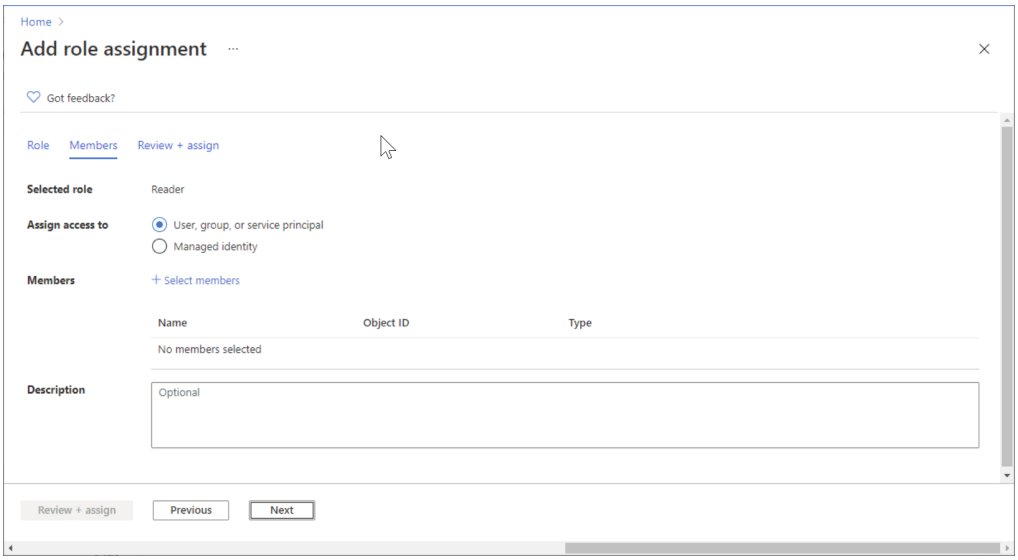
   b. Click the **Role assignments** tab to view the role assignments at this scope.
   c. Click **Add > Add role assignment**.

The **Add role assignment** page opens.



d. On the **Role** tab, select the Key Vault Crypto User role and click **Next**.

e. On the **Members** tab, assign access to **User, group, or service principal**.



f. Under **Members**, click **+ Select members**.

g. Search for `Informatica KMS Connect` and select it.

h. Click **Next**.

i. Review and assign the role assignment.

6. If you use the Vault access policy permission model, perform the following steps to create an access policy for the Informatica KMS Connect app:

   a. In the left panel, select **Access Policies**.

   b. Click **Create** to create a new access policy for the Informatica KMS Connect app.

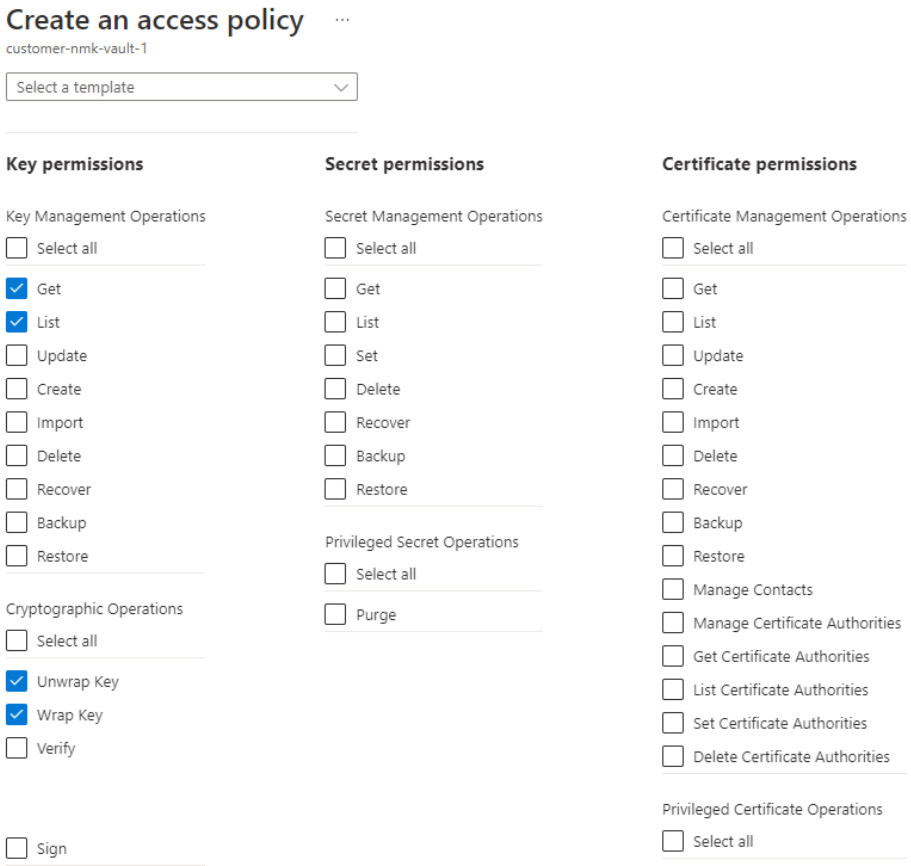
c. Grant the following permissions:

Get

List

Unwrap Key

Wrap Key



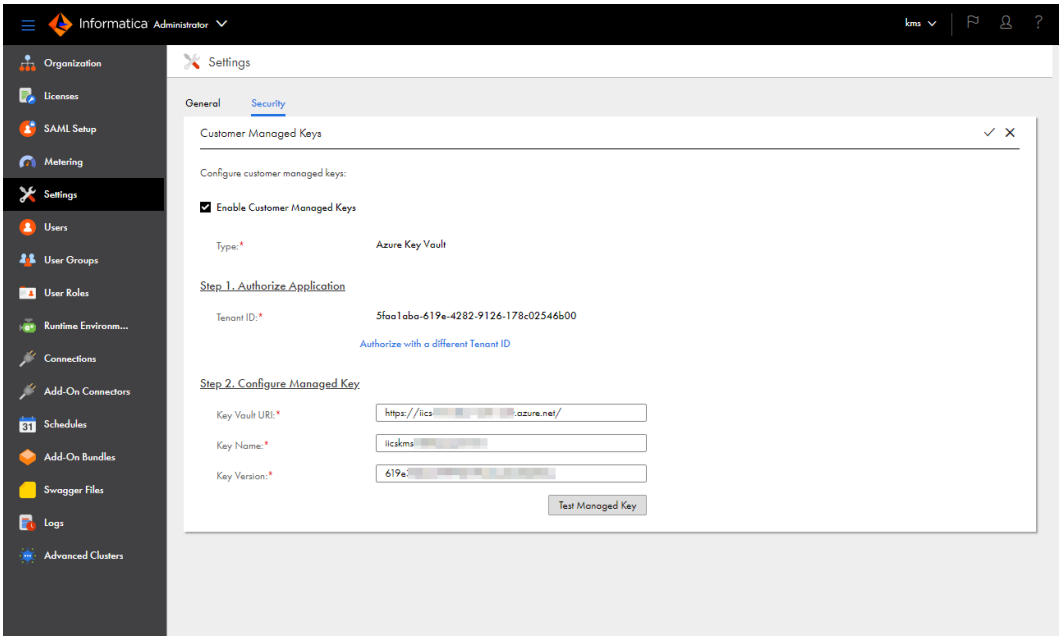
d. Under **Principal**, search for `Informatica KMS Connect` and select it.

e. Click **Next**.

f. Review and create the access policy.

7. Log in to IDMC Administrator with a user account that has both the Admin and Key Admin roles.
8. Open the **Settings** page and click the **Security** tab.
9. Click the edit (pencil) icon.

10. In **Step 2. Configure Managed Key**, enter the **Key Vault URI**, **Key Name**, and **Key Version**:



11. Click **Test Managed Key** to test the key.

    A success message appears if the test was successful.

12. Click the save (check mark) icon to save your changes.

    Note: It can take up to 24 hours for the key to become active.

Frequently asked questions

When I clicked **Test Managed Key** in IDMC, the test failed. What should I do?

If you get an error when testing the key, perform the following checks:

- In IDMC Administrator, verify that the key settings on the **Settings** page match the settings for the CMK in the Azure portal.
- In the Azure portal, verify that the status of the CMK is active.
- In the Azure portal, verify that the permissions on the CMK allow Informatica cryptographic access to the key.

If you continue to encounter errors, contact Informatica Global Customer Support.

What happens if the CMK is rotated in Azure Key Vault?

You can rotate the CMK in Azure Key Vault manually or on a schedule. Rotating a key creates a new version of the key. The old version of the key remains in Azure Key Vault and is used for decryption only.

IDMC automatically detects key rotation. When the CMK is rotated in Azure Key Vault, IDMC decrypts your organization's keys using the old CMK and then encrypts them using the new CMK.

What if I need to update the CMK in Azure Key Vault?

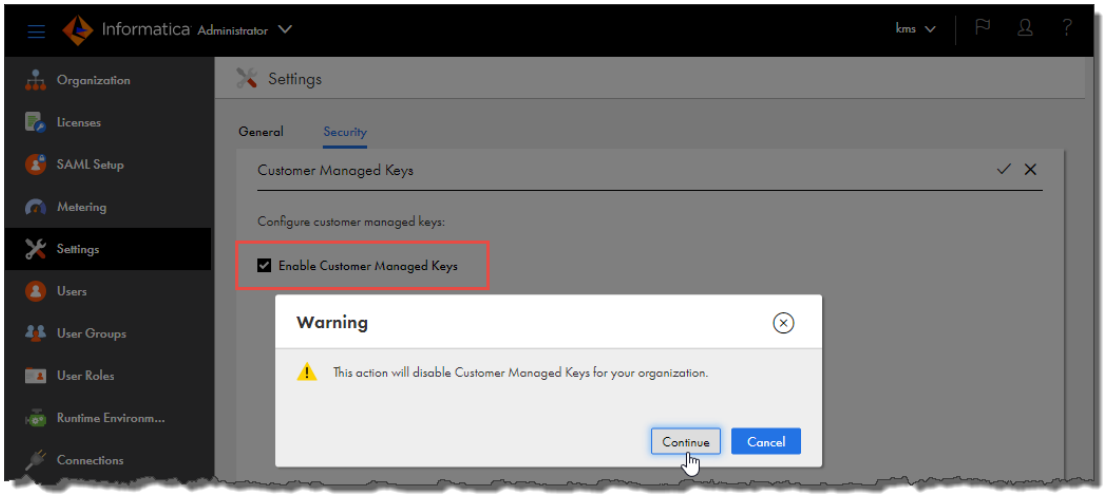
If you need to update the CMK, first provision a new CMK in Azure Key Vault. Then, update the key details on the **Settings** page in IDMC Administrator.

Note: Be sure to keep the old version of the CMK in Azure Key Vault active until you update the key details in IDMC.

You can delete the old version of the CMK in Azure Key Vault after you update the key details on the **Settings** page in IDMC Administrator.

What if I want Informatica to manage key encryption?

If you want Informatica to manage key encryption, you can disable the **Enable Customer Managed Keys** option on the **Settings** page in IDMC Administrator:



When you do this, be sure to keep the current version of the CMK in Azure Key Vault active. If the CMK is not active, disabling customer managed keys in IDMC fails.

When you disable this option, your organization's encryption keys are once again encrypted using encryption keys that are managed by Informatica. It can take up to 10 minutes for the Informatica encryption keys to become active.

You can disable or delete the CMK in Azure Key Vault after you disable the **Enable Customer Managed Keys** option in Administrator.

What if I want to temporarily revoke Informatica's access to the CMK?

If you want to temporarily revoke Informatica's access to the CMK, you can disable the key in Azure Key Vault.

When you disable the CMK, IDMC can no longer unencrypt your organization's encrypted data, and any jobs that use the data will fail until you reactivate the CMK in Azure Key Vault.

How do I replace the CMK if I suspect it has been compromised?

If you want to replace the CMK, you can delete the key in Azure Key Vault and create a new one.

Warning: Deleting the CMK in Azure Key Vault results in permanent loss to any encrypted data in IDMC and causes the jobs that use the data to fail.

If you need to replace the CMK, perform the following steps so that you don't lose access to the encrypted data and jobs don't fail:

1. In Administrator, open the **Settings** page, click the **Security** tab, and disable the **Enable Customer Managed Keys** option.
2. In the Azure portal, delete the CMK.

3. In the Azure portal, create a new CMK.
4. On the **Settings** page in IDMC Administrator, re-enable the **Enable Customer Managed Keys** option and enter the details for the new CMK.

Can I delete the CMK if I don't want Informatica to access any of my encrypted data?

Warning: Deleting the CMK in Azure Key Vault results in permanent loss to any encrypted data in IDMC and causes the jobs that use the data to fail.

If you're sure that you want Informatica to forgo all access to your encrypted data in IDMC, you can delete the CMK in Azure Key Vault.

Author

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